

Nidhi Parab

+1 (716) 970-9624 | Portfolio | nidhiparabmail@gmail.com | [LinkedIn](#) | [GitHub](#)

EDUCATION

University at Buffalo <i>Masters of Science in Computer Science and Engineering (3.6/4.0 GPA)</i>	Buffalo, NY <i>Aug. 2024 – Dec. 2025</i>
---	---

EXPERIENCE

AI/ML R&D and Product Intern <i>Crypt0nest.io LLC</i> - Built time-series forecasting models for digital asset returns using neural networks, achieving 15% higher prediction accuracy than baseline models through advanced hyperparameter tuning. - Implemented AI explainability frameworks (SHAP/LIME) to enhance model transparency, reducing model interpretation time by 60% for regulatory compliance. - Engineered backtesting systems processing 10M+ data points daily, contributing to 95% model reliability in production environments. - Prototyped real-time metric dashboards by streaming model insights via FastAPI to React, experimenting with D3.js and Plotly.js to explore interactive data visualizations.	Jun. 2025 – Present <i>Austin, TX</i>
Undergraduate Research Student <i>Indian Institute of Technology (IIT), Bombay</i> - Built end-to-end deep learning pipelines using TensorFlow, achieving 99.86% accuracy on hyperspectral image classification. - Improved generalization on limited data using Self-Attention and Temperature Scaling; conducted full ablation studies and benchmarks. - Developed custom visualization and explainability tools for high-dimensional model outputs. - Conducted high-impact research recognized at the institutional level; research and code officially copyrighted by the Government of India under IIT Bombay’s research IP framework.	Jun. 2023 – Jun. 2024 <i>Mumbai, India</i>
SDE Intern <i>Indian Development Foundation</i> - Built a full-stack platform using React (Next.js) and MySQL for 300+ nonprofit schools. - Integrated REST APIs to manage grading, teacher data, and school operations in a centralized dashboard. - Enabled staff to access and update data with optimized queries and seamless front-end experience.	Dec. 2022 – Mar. 2023 <i>Mumbai, India</i>

PROJECTS

Colab Alarm <i>Python, Google Colab, PyPI</i> - Published a Python package on PyPI to play customizable audio alerts in Google Colab, improving workflow for ML researchers. - Used by 1K+ developers; built for minimal setup with <code>pip install colab-alarm</code> and a one-line function call.	Apr. 2025
HVAC Maintenance Management System <i>Django, PostgreSQL, Vue.js, CircleCI</i> - Built a full-stack CMMS app to manage HVAC maintenance logs and tasks, aligning with ASHRAE 180 standards. - Integrated CI/CD via CircleCI, and designed a Vue.js frontend for technician task tracking in real time.	Apr. 2025
Buffalo Code Violations Dashboard <i>PostgreSQL, Tableau</i> - Engineered a PostgreSQL database to track and analyze Buffalo city code violations. - Created Tableau dashboards to reveal trends and hotspots for improved urban compliance strategies.	Feb. 2025
GitHub Issue Finder CLI <i>Rust</i> - Built a Rust-based CLI tool using GitHub API to fetch good-first-issues for contributors, supporting async operations with Tokio. - Handled robust JSON parsing and error handling with <code>request</code> and <code>serde</code>, enhancing usability for new devs.	Oct. 2024

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, SQL, Rust
Frameworks/Tools: Django, React, Node.js, Flask, PostgreSQL, MySQL, CircleCI, TensorFlow, Keras
Libraries: Pandas, NumPy, Matplotlib, OpenCV, MediaPipe, SHAP, LIME
Cloud & DevOps: Google Cloud Platform, CI/CD, Docker, Git, Slack, Trello

PUBLICATION

Personalized AI Based Yoga-Pose Recommendation System Using Game Theory: Achieved the ‘Best Paper Award’ in the Artificial Intelligence Track among 120 participants at the 3rd International Conference on Advanced Computing Technologies and Applications (October 2023).

LEADERSHIP & COMMUNITY IMPACT

Mentor , at AnitaB.org UK, 2025. Mentoring 20+ students on ML fundamentals, Python programming, and GitHub workflows using premium resources like Aurélien Géron’s <i>Hands-On Machine Learning</i> .
Microsoft Student Learn Ambassador , 2023-24. Led 2 workshops as a key advocate for Microsoft technologies, fostering a community of learners.
General Secretary at the Computer Society of India (CSI), 2023-24. Overseeing tasks for all the events and coordinating the work of 32 committee members.